

11.16 Classwork: Distributions — Markscheme

1. [Maximum mark: 6] *No calculator.*

The binomial expansion of $(1 + kx)^n$ is given by $1 + \frac{9x}{2} + 15k^2x^2 + \dots + k^nx^n$, where $n \in \mathbb{Z}^+$ and $k \in \mathbb{Q}$. Find n and k . [6]

Apply the binomial expansion. **M1**

$$(1 + kx)^n = 1 + \binom{n}{1}kx + \binom{n}{2}k^2x^2 + \dots$$

Coefficient of x : $nk = \frac{9}{2}$ **A1**

Coefficient of x^2 : $\binom{n}{2}k^2 = 15k^2 \Rightarrow \binom{n}{2} = 15$ **A1**

$$\frac{n(n-1)}{2} = 15 \Rightarrow n^2 - n - 30 = 0$$

Valid attempt to solve. **M1**

$(n-6)(n+5) = 0$, and $n \in \mathbb{Z}^+$, so $n = 6$. **A1**

Substitute back: $6k = \frac{9}{2}$, so $k = \frac{3}{4}$. **A1**

2. [Maximum mark: 4]

Find the coefficient of x^6 in the expansion of $(2x - 5)^9$. [4]

General term: $\binom{9}{r}(2x)^{9-r}(-5)^r$. **M1**

For x^6 , need $9 - r = 6$, so $r = 3$.

Term: $\binom{9}{3}(2x)^6(-5)^3$. **A1**

$\binom{9}{3} = 84$, $(2)^6 = 64$, $(-5)^3 = -125$. **A1**

Coefficient = $84 \times 64 \times (-125) = -672000$. **A1**

Note: Award A0 for final answer $-672000x^6$.

3. [Maximum mark: 7]

(a) Probability that exactly two semiconductors are defective. [4]

Recognition of binomial distribution; let $X \sim B(10, p)$. **M1**

GDC: binomPdf(10, p, 2) for each machine.

For machine A ($p = 0.10$): $P(X = 2) = \binom{10}{2}(0.10)^2(0.90)^8 = 0.193710\dots$

For machine B ($p = 0.05$): $P(X = 2) = \binom{10}{2}(0.05)^2(0.95)^8 = 0.0746347\dots$

Attempt to consider both machines using $p = 0.1$ **AND** $p = 0.05$. **M1**

$P(2 \text{ defective}) = 0.5(0.193710\dots) + 0.5(0.0746347\dots)$ **A1**

$= 0.134172\dots = 0.134$ **A1**

(b) Given exactly two defective, find $P(\text{machine A})$. [3]

Recognition of conditional probability. **M1**

$P(A \mid 2 \text{ def}) = \frac{P(A \cap 2 \text{ def})}{P(2 \text{ def})} = \frac{0.5 \times 0.193710\dots}{0.134172\dots} = \frac{0.0968\dots}{0.1342\dots}$ **A1**

$= 0.72187\dots = 0.722$ **A1**

Note: Accept 0.723 or 0.724 from using 3 s.f. values from part (a).

4. [Maximum mark: 5]

$D \sim N(32, \sigma^2)$, IQR = 0.28. Find σ . [5]

METHOD 1 (using quartile values).

By symmetry, $Q_1 = 32 - 0.14 = 31.86$ or $Q_3 = 32 + 0.14 = 32.14$. **A1**

Recognition that $P(D < Q_1) = 0.25$ (or $P(D > Q_3) = 0.25$). **M1**

z -score for the lower quartile: $z = \text{invNorm}(0.25) = -0.674489\dots$ **A1**

GDC: invNorm(0.25, 0, 1).

Standardise: $\frac{31.86 - 32}{\sigma} = -0.674489\dots$ **A1**

$\sigma = \frac{0.14}{0.674489\dots} = 0.207564\dots = 0.208 \text{ (mm)}$ **A1**

METHOD 2 (using IQR directly).

Recognition that IQR = $Q_3 - Q_1 = 2 \times 0.674489\dots \times \sigma$ (areas 0.25 each side). **M1**

$z = \pm 0.674489\dots$ **A1**

$2 \times 0.674489 \times \sigma = 0.28$ **A1A1**

$\sigma = 0.207564\dots = 0.208 \text{ (mm)}$ **A1**

5. [Maximum mark: 8]

(a) Estimate arm span for adult with foot length 19.8 cm. [2]

Substitute $F = 19.8$ into the regression line for A on F (we are predicting A). **M1**

$$A = 2.89(19.8) + 99.3 = 156.522 \text{ cm}$$

Arm span ≈ 157 cm.

A1

Note: Award M0A0 if candidate uses F on A regression instead, giving $A \approx 156$.

(b) Find mean arm span and mean foot length.

[3]

Recognition that the two regression lines intersect at the mean point $(\bar{A}, \bar{F})$.

M1

GDC: solve the system $F = 0.335A - 32.6$ and $A = 2.89F + 99.3$.

Substituting: $A = 2.89(0.335A - 32.6) + 99.3$

$$A(1 - 0.96815) = -94.214 + 99.3$$

$$\bar{A} = 159.686 \dots \approx 160 \text{ cm}$$

A1

$$\bar{F} = 0.335(159.686) - 32.6 = 20.8948 \dots \approx 20.9 \text{ cm}$$

A1

(c) Find σ for $H \sim N(163, \sigma^2)$, given $P(153 < H < 173) = 0.88$.

[3]

By symmetry of the normal curve about 163, the tails total 0.12, so $P(H < 153) = 0.06$ (or $P(H < 173) = 0.94$).

M1

z -score: $z = \text{invNorm}(0.06) = -1.55477 \dots$

A1

GDC: $\text{invNorm}(0.06, 0, 1)$.

Standardise: $\frac{153 - 163}{\sigma} = -1.55477$, so $\sigma = \frac{10}{1.55477 \dots}$

$$\sigma = 6.43181 \dots = 6.43 \text{ cm}$$

A1

Note: Accept METHOD 2 — solve $\text{normCdf}(153, 173, 163, \sigma) = 0.88$ on the GDC.